

Supply Chain Analytics: Phase 1 – Data Validation & Cleaning

Project Overview

This phase focuses on establishing a **"Single Source of Truth"** by auditing the raw supply chain database. We identify structural inconsistencies, validate relational integrity, and develop a cleaning layer using SQL views to prepare the data for advanced risk modeling.

1. Data Audit & Integrity Checks

Before transformation, we performed a rigorous audit of the raw schema to identify "data leakage" and orphaned records.

Row Count Baseline

established initial baselines across all core tables to ensure data ingestion was complete:

- **Entities:** Customers, Products, Suppliers, Warehouses.
- **Transactions:** Purchase Orders, PO Items, Shipments, Inventory Levels.

A professional README for a data analysis project should bridge the gap between technical implementation and business value. This Phase 1 README focuses on **Data Integrity and Structural Cleaning**.

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Relational Integrity (Foreign Key Validation)

We conducted `left join` audits to identify orphaned records. These are critical "silent killers" in supply chain data that lead to incorrect inventory totals.

- **Inventory Audit:** Verified that every inventory record maps to a valid warehouse and product.
- **Procurement Audit:** Verified that all POs and PO Items are linked to active suppliers and products.
- **Logistics Audit:** Checked for shipments tied to non-existent POs or warehouses.

2. Transformation & Cleaning Layer (Views)

Rather than altering raw data, we implemented a **View-based Cleaning Layer**. This ensures that the original data remains intact while providing sanitized data for analysis.

`purchase_order_cn`

- **Standardization:** Normalized order statuses to lowercase/trimmed strings.
- **Date Sanity:** Flagged records where `expected_delivery` preceded the `order_date`.
- **Financial Scrubbing:** Rounded amounts and handled NULL/Zero totals.

`vw_shipments_clean`

- **Carrier Normalization:** Standardized carrier names to prevent duplication in reports (e.g., "UPS" vs "ups").
- **Cancellation Logic:** Automatically flags shipments tied to canceled POs to prevent "ghost inventory" projections.

inventory_cn

- **Dynamic Risk Flagging:** Created the `is_below_reorder_level` logic by comparing `live_quantity_on_hand` against the product `reorder_level`.
- **Constraint Enforcement:** Excluded negative inventory counts that represent data entry errors.

vw_products_clean & vw_suppliers_clean

- **Deduplication:** Utilized `DISTINCT` logic to ensure product-level analysis isn't skewed.
- **Standardization:** Normalized order statuses to trimmed strings (e.g product name,description,SKU)
- **Status Mapping:** Standardized "Suspended" or "Active" labels for suppliers to enable risk filtering.
- **Flagging numerical anomalies :** (e.g `p.reorder_level < 0`, `p.unit_price IS NULL` OR `p.unit_price <= 0`)

3. Preliminary Findings

Data Quality & Integrity Issues Identified

- **Orphaned Records Detected**
 - **2** purchase order records were excluded due to missing or invalid references.
 - **14** shipment records were excluded because they were not linked to a valid purchase order ID or warehouse ID.
 - **24** invalid records were identified in the purchase order items table.
 - **64** records were excluded from the inventory table due to data integrity issues.
- **Reference Data Validation**
 - No records were removed from the **suppliers** or **products** tables.
 - However, **suspended suppliers** and **suspicious product records** were flagged for monitoring rather than excluded, to preserve analytical visibility.
- **Date Consistency Issues**

- **38** records with invalid or inconsistent date values were identified and flagged for downstream handling.
- **Supplier Risk Indicators**
 - **7** suppliers with a suspended status were flagged to track ongoing transactional activity and potential operational risk.
- **Inventory Risk Signals**
 - **5** critical stockout cases were detected, indicating immediate replenishment or supplier attention is required.

4. Next Steps

With the data sanitized and the relational integrity verified, the project will move to **Phase 2: Performance Metrics**, where we will calculate:

1. Lead Time Variability.
2. Supplier On-Time Delivery (OTD) rates.
3. Carrier reliability scores.